Unit 1, Lesson 12: Angle Bisectors

Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

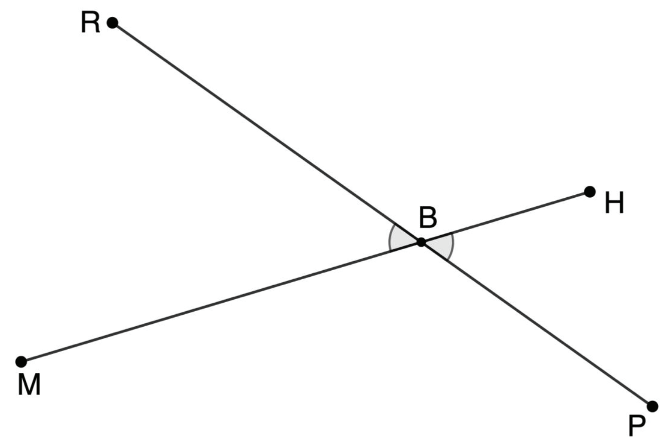
MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.

Learning Target

☐ I can construct an angle bisector.

1.12.1 Warm-Up: Agree or Disagree

1. Line segments MH and RP intersect at point B.



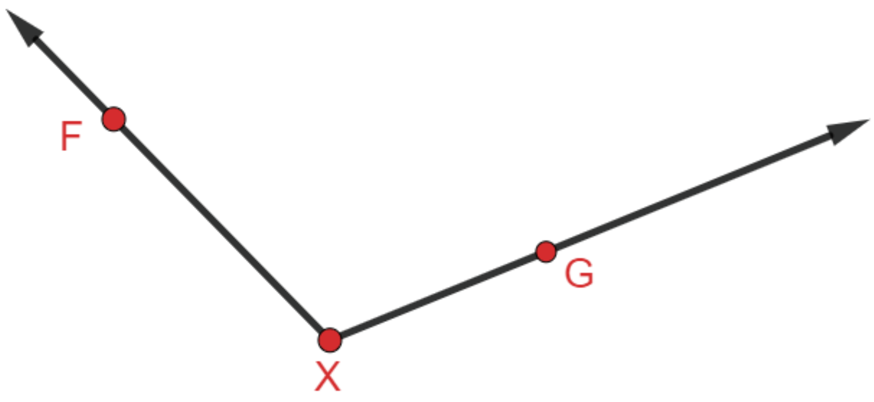
   a. Jamessia says $\angle MBR$ and $\angle PBH$ have the same measure. Malik disagrees and says that obviously the measure of $\angle MBR$ is larger.

      Where do you stand?
      - ○ I agree with Jamessia.
      - ○ I agree with Malik.
      - ○ I disagree with both students.

   b. Explain your choice for Part A.

1.12.2 Collaboration: Bisecting an Angle

1. $\angle FXG$ is shown.

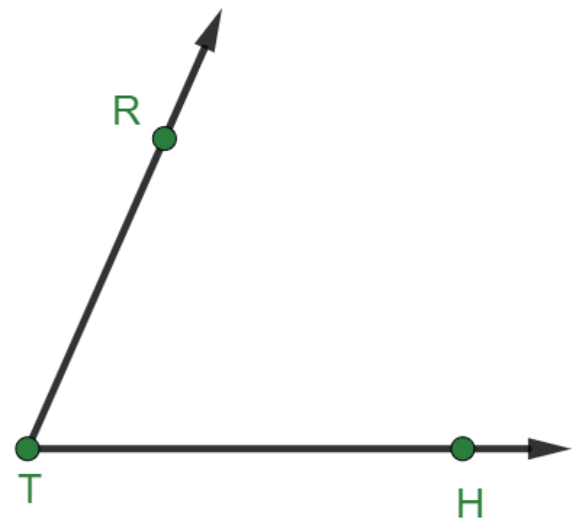


   a. Measure the angle with a protractor.

   b. Draw $\overrightarrow{XY}$ that bisects $\angle FXG$.

   c. Discuss with your partner why bisecting $\angle FXG$ using a protractor and a ruler may not be accurate. Summarize your discussion.

2. $\angle RTH$ is shown.



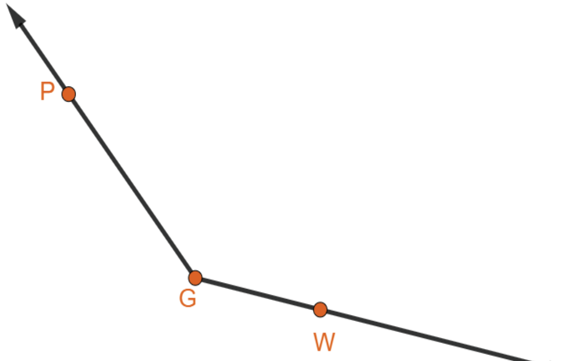
   a. Use patty paper to copy $\angle RTH$.

   b. Fold the patty paper so that $\overrightarrow{TR}$ and $\overrightarrow{TH}$ lie on each other. Open the paper and draw $\overrightarrow{TM}$ along the crease.

   c. Explain how folding the patty paper bisects the angle.

Another way to more accurately bisect an angle is to use a compass and a ruler.

3. $\angle PGW$ is shown.



Work with your partner and use the steps below to bisect $\angle PGW$.

- Place the compass' point on point G.
- Adjust the compass' width to any width that will stay on the angle.
- Use the setting to draw an arc that extends across each leg of $\angle PGW$. Label the intersections of the arc and the two sides of the angle points F and H.
- Place the compass' point on point F. Keep the width of the compass the same or open the compass wider than the width used to draw the arc in the previous step. Draw an arc in the interior of the angle.
- Without changing the compass' setting, place the compass' point on point H and make an arc in the interior of the angle. The two small arcs in the interior of $\angle PGW$ should be intersecting. If they are not intersecting, adjust the compass to a larger width and draw new arcs from points F and H.
- Draw and label point Y where the two arcs intersect.
- Use a straightedge to draw a ray from point G through point Y.
- $\overrightarrow{GY}$ bisects $\angle PGW$.

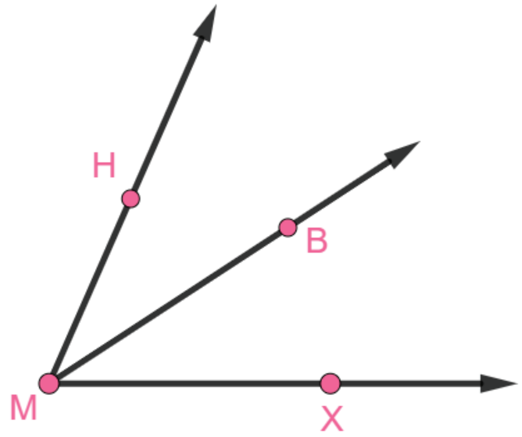
4. Olivia found some steps for bisecting an angle on the internet. She wrote them down and then mixed them up to test herself. Her mixed-up steps are shown. Work with your partner to order the steps 1-9.

	Draw a ray from the vertex of the angle to the intersection of the two small arcs.
	Stretch the compass to any length that stays on the angle.
	Without changing the span on the compass, place the point of the compass on the other point made by the arc and the other side of the angle and make a similar arc.
	Place the compass point on one of the two new points made by the arc.
	Change the compass width so that the pencil stays between the rays of the angle.
	Place compass point on the vertex of the angle.
	Swing the compass to make an arc that crosses both rays of the angle.
	Draw an arc in the interior of the angle.
	Mark the point where the two small arcs in the interior of the angle intersect.

1.12.3 Guided Instruction: Using Angle Bisectors to Solve Problems

An ***angle bisector*** is a line, ray, or line segment that cuts an angle into two equal parts.

1. $\angle HMX$ is shown where $\overrightarrow{MB}$ bisects $\angle HMX$.



If $m\angle HMX = (7x + 3)°$ and $m\angle BMX = (3x + 6)°$, then find the value of x.

2. $\overrightarrow{DY}$ bisects $\angle WDN$ such that $m\angle YDN = (10x - 4)°$ and $m\angle WDN = (17x + 16.75)°$.

   a. Draw a diagram of the description.

   b. Yordis and Hae-Won used the given information to determine the value of x. Their equations are shown.

Yordis	Hae-Won
$10x - 4 = \frac{1}{2}(17x + 16.75)$	$\frac{1}{2}(10x - 4) = 17x + 16.75$

   c. Discuss with your partner which student made a mistake setting up the equations.

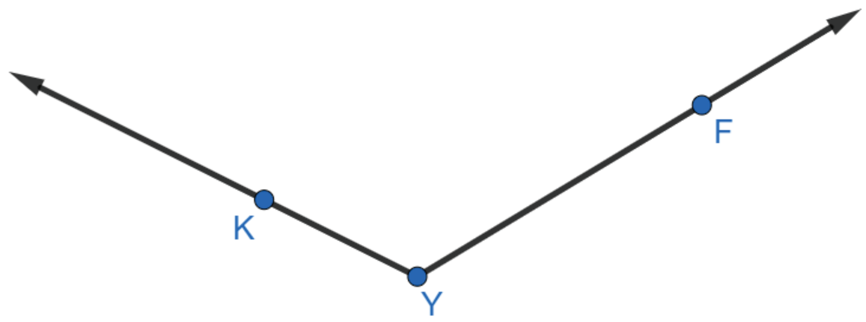
   d. Write a short note to the student who made a mistake in setting up the equations to explain their error to them.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

3. $\overrightarrow{BT}$ bisects $\angle HBN$ such that $m\angle HBN = (10x - 2)°$, $m\angle HBT = (5x - 1)°$, and $m\angle TBN = (7x - 23)°$. Find $m\angle TBN$.

1.12.4 Wrap-Up: Ticket Out the Door

1. Construct the angle bisector of $\angle KYF$ using a compass and a straightedge.



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